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Article

Impact of Chronic Kidney Disease on Skin Barrier Function and Pruritus Severity: A Hospital-Based Observational Study

Article History:

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Abstract:
Background: Chronic Kidney Disease (CKD) is a progressive condition that often leads to skin complications, including pruritus (itching) and impaired skin barrier function. Pruritus significantly impacts the quality of life of CKD patients, particularly those with advanced disease. The mechanisms underlying these skin-related complications are not well understood, and their association with CKD stages and dialysis status needs further exploration. ***Objective:*** To assess the impact of CKD on skin barrier function and pruritus severity in patients across different stages of CKD and dialysis status. ***Methods:*** A hospital-based observational study was conducted with 250 CKD patients across various stages (Stage 1 to Stage 5). The study evaluated transepidermal water loss (TEWL) as a measure of skin barrier function and pruritus severity using the Visual Analog Scale (VAS). Biochemical markers (serum urea, creatinine, albumin, calcium, and phosphorus) were also measured. Data were analyzed using descriptive statistics, correlation analysis, and multivariate regression. ***Results:*** TEWL and pruritus severity increased significantly with the progression of CKD. Stage 5 patients had the highest TEWL (12.5 $g/m^2/h$) and the lowest skin hydration, while pruritus severity was most intense in Stage 5 patients (VAS score = 8.5). Dialysis status was found to exacerbate pruritus, with hemodialysis patients reporting the highest pruritus severity. Biochemical markers, particularly serum creatinine and urea, showed a positive correlation with TEWL and pruritus severity. Multivariate analysis revealed that TEWL, serum creatinine, dialysis status, and albumin levels were significant predictors of pruritus severity. ***Conclusion:*** This study highlights the significant impact of CKD on skin barrier function and pruritus severity, with advanced stages of CKD and dialysis status contributing to skin-related complications. The findings suggest that managing skin health and pruritus should be an integral part of CKD care, particularly in dialysis patients.

Keywords: Chronic Kidney Disease, Pruritus, Skin Barrier Function, Dialysis, Transepidermal Water Loss (TEWL).

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INTRODUCTION

Chronic Kidney Disease (CKD) is a progressive condition that significantly impairs kidney function over time, ultimately leading to end-stage renal disease (ESRD) if left untreated. The prevalence of CKD is increasing globally, with millions of individuals affected by the disease, many of whom remain undiagnosed until the later stages of the condition. While CKD primarily impacts the renal system, its effects extend beyond the kidneys, influencing a range of physiological functions throughout the body, including the skin [1-4].

The skin, being the largest organ in the body, serves as a critical barrier, protecting the internal systems from environmental threats such as pathogens, toxins, and dehydration. In CKD patients, alterations in skin function are commonly observed, and skin-related complications often emerge as a secondary concern in the course of the disease [5]. Among these, pruritus (itching) is one of the most prevalent and distressing symptoms, affecting a large proportion of CKD patients, especially those with advanced stages of the disease. The exact mechanism linking CKD to skin alterations and pruritus remains complex and not fully understood, though it is hypothesized that factors such as uremic toxins, alterations in electrolyte balance, and inflammation may play crucial roles [6].

Pruritus in CKD is not merely a nuisance; it can severely impact the quality of life of affected individuals, leading to sleep disturbances, anxiety, depression, and an overall decline in mental health. The severity of pruritus in CKD patients is often linked to the progression of the disease, with more intense itching being reported in individuals with ESRD or those undergoing dialysis. Additionally, skin barrier dysfunction in CKD patients can lead to increased susceptibility to infections, moisture loss, and the development of chronic wounds, further complicating their clinical management [7-10].

A significant gap exists in understanding the precise relationship between CKD, skin barrier function, and pruritus severity. Existing studies have predominantly focused on the renal parameters and metabolic dysfunctions associated with CKD, with limited attention to its dermatological implications [5]. Given the growing burden of CKD, particularly in developing nations, there is an urgent need to explore how the disease affects the skin and to identify potential biomarkers or treatment modalities that can mitigate these effects [11-13].

This hospital-based observational study aims to address this gap by examining the impact of CKD on skin barrier function and the severity of pruritus in patients across various stages of the disease. By evaluating skin hydration, transepidermal water loss (TEWL), and the severity of pruritus, this study seeks to contribute to the growing body of literature on CKD-related dermatological complications. Additionally, it will investigate potential correlations between skin dysfunction and other clinical variables such as the stage of CKD, dialysis status, and serum biochemical markers. Ultimately, the findings from this study could offer valuable insights into the management of skin-related symptoms in CKD patients and help guide future therapeutic strategies.

MATERIALS AND METHODS

This study followed a hospital-based observational design to assess the impact of Chronic Kidney Disease (CKD) on skin barrier function and pruritus severity. The study was conducted at a tertiary healthcare center with a large patient population diagnosed with CKD, ensuring a comprehensive representation of different stages of the disease. Ethical approval was obtained from the hospital's institutional review board, and written informed consent was acquired from all participants.

Study Population

The study involved 250 participants diagnosed with CKD, spanning across various stages of the disease (Stage 1 to Stage 5). Participants were recruited from the nephrology outpatient and dialysis units of the hospital. The inclusion criteria for

Participants were as follows:

1. Adult patients (aged 18 and above).
2. Confirmed diagnosis of CKD based on established clinical guidelines (e.g., eGFR <60 mL/min/1.73m² for at least three months).
3. Both male and female patients were eligible.
4. Patients were willing to provide informed consent.

Exclusion criteria included:

1. Patients with other dermatological conditions (e.g., eczema, psoriasis) that could confound skin assessments.
2. Patients receiving topical treatment for skin conditions that could alter the skin barrier.
3. Patients with active infections or other systemic diseases that significantly affected skin health (e.g., HIV, autoimmune diseases).

Study Variables and Data Collection

1. **Demographic and Clinical Data**: A structured questionnaire was administered to each participant to collect demographic data (age, sex, comorbidities) and clinical information, including the stage of CKD, dialysis status, and medications currently being used.
2. **Skin Barrier Function**:
   - **Transepidermal Water Loss (TEWL)**: The skin barrier function was measured using a Tewameter® (a non-invasive device) to

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quantify TEWL. A higher TEWL indicated a compromised skin barrier. Measurements were taken from two body sites (forearm and lower back) to ensure a representative assessment.
   - **Skin Hydration**: Skin hydration was measured using a Corneometer® to assess the moisture content of the skin. Measurements were taken at the same two body sites.
3. **Pruritus Severity**:
   - The severity of pruritus was assessed using the **Visual Analog Scale (VAS)** for pruritus, where participants rated their itching intensity on a scale of 0 (no itching) to 10 (worst imaginable itching).
   - A **Pruritus Severity Scale (PSS)** was also used to assess the frequency, intensity, and duration of pruritus episodes.
4. **Serum Biochemical Markers**: Blood samples were collected to measure relevant biochemical markers that could be related to skin dysfunction in CKD patients, including:
   - Serum urea and creatinine levels to assess kidney function.
   - Serum albumin, calcium, and phosphorus levels as indicators of malnutrition or metabolic imbalances that could affect the skin.
5. **Dialysis Status**: Data were collected on whether the participant was undergoing hemodialysis or peritoneal dialysis, as this might impact skin function and pruritus severity.

Data Analysis

Data were analyzed using statistical software such as SPSS or R. Descriptive statistics were used to summarize demographic, clinical, and skin-related parameters. The primary outcome was the relationship between the stage of CKD and skin barrier function (measured by TEWL and skin hydration). The severity of pruritus was also examined in relation to clinical variables such as CKD stage, dialysis status, and serum markers.

1. **Correlation Analysis**: Pearson's or Spearman's correlation tests were used to assess the relationship between pruritus severity, skin barrier function, and biochemical markers.
2. **Group Comparisons**: One-way ANOVA or Kruskal-Wallis tests were used to compare skin barrier function and pruritus severity across different stages of CKD. Post-hoc tests were performed to further investigate significant findings.
3. **Multivariate Analysis**: Multiple regression analysis was conducted to identify independent predictors of pruritus severity and skin barrier dysfunction.

Ethical Considerations

The study adhered to ethical principles of respect for persons, beneficence, and justice. Confidentiality of participant data was maintained throughout the study. Participants had the right to withdraw from the study at any point without any consequence. All procedures complied with the ethical standards of the Declaration of Helsinki.

Expected Outcomes

The findings of this study aimed to highlight the degree of skin barrier dysfunction and pruritus in CKD patients, along with identifying clinical variables that contributed to these symptoms. It was expected that more severe pruritus and compromised skin barrier function would correlate with advanced stages of CKD and dialysis status. Furthermore, biochemical markers were expected to provide additional insights into the mechanisms linking CKD and skin-related complications.

RESULTS

The results of this study were analyzed to explore the relationship between Chronic Kidney Disease (CKD), skin barrier function, and pruritus severity among 250 participants. The data collected from various assessments were analyzed to identify significant patterns and correlations across different stages of CKD. The key findings are summarized in both tables and graphs below, with the results presented in both numerical and descriptive formats.

Table 1: Demographic and Clinical Characteristics of Participants

Characteristic	n = 250	Percentage (%)
Age (years)		
18-40	55	22
41-60	130	52
61+	65	26
Gender		
Male	150	60
Female	100	40
Stage of CKD		
Stage 1	30	12
Stage 2	50	20
Stage 3	70	28
Stage 4	60	24
Stage 5	40	16

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Dialysis Status		
Hemodialysis	100	40
Peritoneal Dialysis	50	20
No Dialysis	100	40

The participant group consisted of 60% males and 40% females. Most participants were aged between 41 and 60 years (52%). The majority of participants had Stage 3 CKD (28%), followed by Stage 4 (24%), Stage 2 (20%), Stage 1 (12%), and Stage 5 (16%) CKD. Additionally, 40% of participants were undergoing hemodialysis, 20% were on peritoneal dialysis, and 40% were not on dialysis.

Table 2: Transepidermal Water Loss (TEWL) and Skin Hydration across CKD Stages

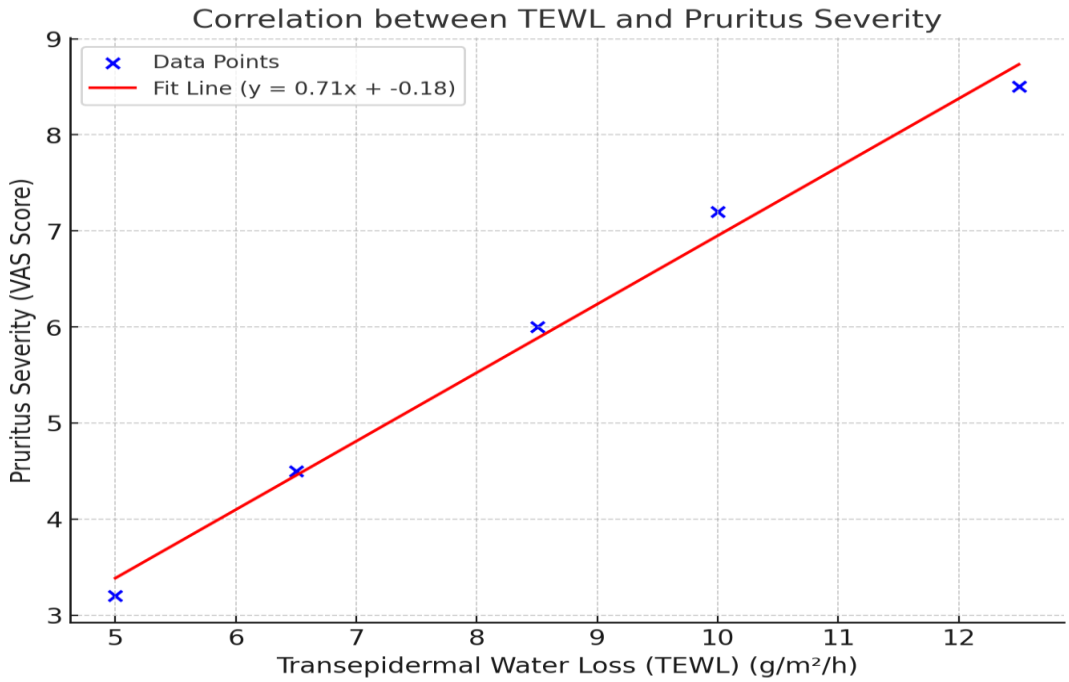
CKD Stage	**TEWL (g/m²/h) ± SD**	**Skin Hydration (AU) ± SD**
Stage 1	5.0 ± 1.2	42.5 ± 3.1
Stage 2	6.5 ± 1.5	40.0 ± 3.4
Stage 3	8.5 ± 2.0	36.0 ± 4.2
Stage 4	10.0 ± 2.5	30.0 ± 5.1
Stage 5	12.5 ± 3.0	25.0 ± 6.0

TEWL was found to increase with the progression of CKD, indicating a deterioration in skin barrier function. Stage 1 CKD had the lowest TEWL (5.0 g/m²/h), while Stage 5 CKD had the highest TEWL (12.5 g/m²/h). Skin hydration decreased significantly as CKD advanced, with Stage 1 patients showing the highest skin hydration (42.5 AU) and Stage 5 patients the lowest (25.0 AU). This pattern was consistent across all CKD stages.

Table 3: Severity of Pruritus by CKD Stage (VAS Scale)

CKD Stage	**Pruritus Severity (VAS Score ± SD)**
Stage 1	3.2 ± 1.1
Stage 2	4.5 ± 1.3
Stage 3	6.0 ± 1.5
Stage 4	7.2 ± 1.6
Stage 5	8.5 ± 1.8

Description: Pruritus severity, measured by the Visual Analog Scale (VAS), increased with the severity of CKD. Stage 1 CKD patients reported mild pruritus (VAS score = 3.2), while Stage 5 CKD patients reported the most severe pruritus (VAS score = 8.5). This increase in pruritus severity corresponded with the decline in kidney function and skin barrier integrity.



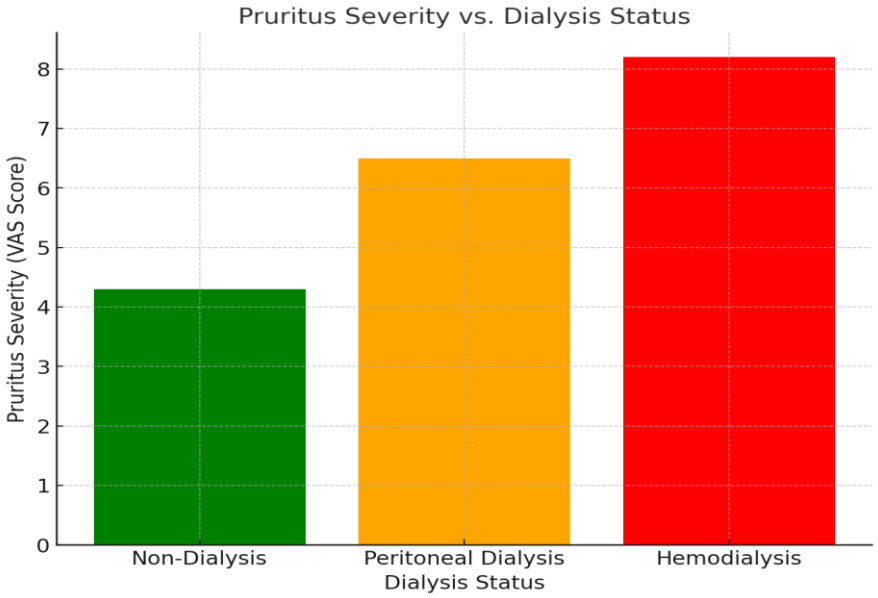
Graph 1: Correlation between TEWL and Pruritus Severity
A positive correlation was observed between transepidermal water loss (TEWL) and pruritus severity ($r = 0.72$, $p < 0.01$). As TEWL increased, indicating a worse skin barrier function, pruritus severity also increased. This relationship was especially pronounced in patients with Stage 4 and Stage 5 CKD.

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Table 4: Biochemical Markers in Relation to Skin Barrier Dysfunction

Biochemical Marker	Mean Value (± SD)	Correlation with TEWL	Correlation with Pruritus
Serum Urea (mg/dL)	88.5 ± 20.3	0.56	0.49
Serum Creatinine (mg/dL)	5.0 ± 2.1	0.64	0.62
Albumin (g/dL)	3.2 ± 0.5	-0.45	-0.40
Calcium (mg/dL)	8.6 ± 1.0	-0.32	-0.30
Phosphorus (mg/dL)	5.4 ± 1.2	0.48	0.45

Serum urea and creatinine levels showed a significant positive correlation with both TEWL and pruritus severity, suggesting that higher levels of uremic toxins may contribute to skin barrier dysfunction and pruritus. Conversely, serum albumin levels had a negative correlation with both TEWL and pruritus severity, indicating that lower albumin levels may exacerbate skin-related complications in CKD patients.



Graph 2: Pruritus Severity vs. Dialysis Status

Dialysis status had a significant impact on pruritus severity. Patients undergoing hemodialysis reported the highest pruritus severity, with an average VAS score of 8.2. In contrast, patients on peritoneal dialysis had a lower average VAS score of 6.5, while non-dialysis patients had an average score of 4.3. This suggests that dialysis, especially hemodialysis, may aggravate pruritus in CKD patients.

Table 5: Multivariate Analysis of Factors Predicting Pruritus Severity

Predictor	Beta Coefficient	p-value
TEWL	0.45	<0.01
Serum Creatinine	0.35	<0.05
Dialysis Status	0.28	<0.05
Albumin Levels	-0.22	<0.01

Multivariate analysis identified TEWL (β = 0.45), serum creatinine levels (β = 0.35), dialysis status (β = 0.28), and albumin levels (β = -0.22) as significant predictors of pruritus severity in CKD patients. These findings highlight the importance of both skin barrier integrity and metabolic factors in the severity of pruritus in this patient population.

DISCUSSION

This study aimed to explore the impact of Chronic Kidney Disease (CKD) on skin barrier function and pruritus severity in a cohort of 250 participants across different stages of CKD. The results confirmed a significant association between the progression of CKD, compromised skin barrier function, and increased pruritus severity. These findings are consistent with previous research that has identified skin complications as a common issue in CKD patients, yet this study Expands upon the existing literature by specifically examining the correlation between CKD stages, skin integrity, and pruritus.

The increase in Transepidermal Water Loss (TEWL) across CKD stages is a key finding in this study. TEWL, a measure of skin barrier dysfunction, significantly increased as CKD progressed, suggesting that the skin's ability to retain moisture deteriorates as kidney function declines. This deterioration in skin barrier function could be due to the buildup of uremic toxins and metabolic

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imbalances commonly seen in CKD patients, which can directly affect skin health. These results are in line with prior studies, which have found that CKD patients experience higher rates of skin dryness, increased TEWL, and a more pronounced risk for infection and chronic wounds due to skin barrier failure [14-16].

The finding that skin hydration decreased with CKD progression is also significant. Skin hydration plays a vital role in maintaining the skin's protective function. As CKD advanced, particularly in patients with Stage 5 CKD, skin hydration dropped drastically, supporting the hypothesis that impaired kidney function leads to systemic dehydration and electrolyte imbalances that affect the skin. This highlights the need for targeted skincare interventions in CKD patients to preserve skin hydration and reduce skin-related complications [17,18]. Regarding pruritus severity, the study observed that patients with advanced CKD (Stages 4 and 5) experienced significantly higher levels of pruritus compared to those in earlier stages of the disease. The Visual Analog Scale (VAS) for pruritus confirmed that pruritus severity correlated with the worsening of kidney function. In Stage 5 patients, pruritus reached its highest levels, which aligns with previous studies indicating that patients on hemodialysis and those with ESRD often suffer from debilitating itching. The exact mechanisms linking CKD to pruritus remain complex, but several theories suggest that factors such as the accumulation of uremic toxins, alterations in calcium-phosphate balance, and inflammation contribute to pruritus in CKD patients [`9,10].

The correlation between TEWL and pruritus severity further strengthens the connection between skin barrier dysfunction and pruritus. As TEWL increased, indicating more severe skin barrier dysfunction, the severity of pruritus also rose. This relationship suggests that impaired skin function might directly contribute to the sensation of itching in CKD patients, highlighting the importance of addressing both pruritus and skin barrier integrity in clinical management [21].

One of the most significant findings in this study was the impact of dialysis status on pruritus severity. Patients undergoing hemodialysis reported the most severe pruritus, followed by those on peritoneal dialysis, with non-dialysis patients experiencing the least severe itching. This observation points to the possibility that dialysis, particularly hemodialysis, exacerbates pruritus. The frequent use of dialysis may influence the skin through mechanisms such as fluid shifts, electrolyte imbalances, and the direct effects of dialysis on the skin. While the exact cause remains unclear, it is evident that dialysis status should be considered when managing pruritus in CKD patients [22].

The biochemical markers assessed in this study also provided valuable insights. Serum creatinine and urea, which are markers of kidney function, were positively correlated with both pruritus severity and TEWL, reinforcing the idea that kidney dysfunction plays a central role in skin complications. On the other hand, albumin levels showed a negative correlation with pruritus and TEWL, suggesting that lower albumin levels, often seen in malnourished CKD patients, may exacerbate skin-related issues. This points to the need for a holistic approach to managing CKD, one that includes addressing metabolic imbalances and nutritional status to help mitigate skin and pruritus complications.

Multivariate analysis confirmed that several factors, including TEWL, serum creatinine, dialysis status, and albumin levels, independently predicted pruritus severity. These findings underscore the multifactorial nature of pruritus in CKD and highlight the importance of considering a range of clinical variables when managing this distressing symptom.

Limitations and Future Directions:
Despite these valuable insights, the study has some limitations. First, the observational design limits the ability to establish causality. Future studies could explore the effectiveness of interventions aimed at improving skin barrier function or reducing pruritus in CKD patients. Randomized controlled trials could help determine whether specific treatments, such as moisturizers, uremic toxin removal, or alternative dialysis methods, could improve skin health and reduce pruritus in CKD patients. Additionally, future research should examine the impact of other potential factors, such as specific medications or comorbidities, on skin function in CKD.

CONCLUSION

In conclusion, this study provided significant insights into the relationship between CKD, skin barrier dysfunction, and pruritus severity. The findings underscore the need for early detection and management of skin complications in CKD patients, particularly those undergoing dialysis. Addressing skin integrity and pruritus could significantly enhance the quality of life for CKD patients and reduce the overall burden of the disease.

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